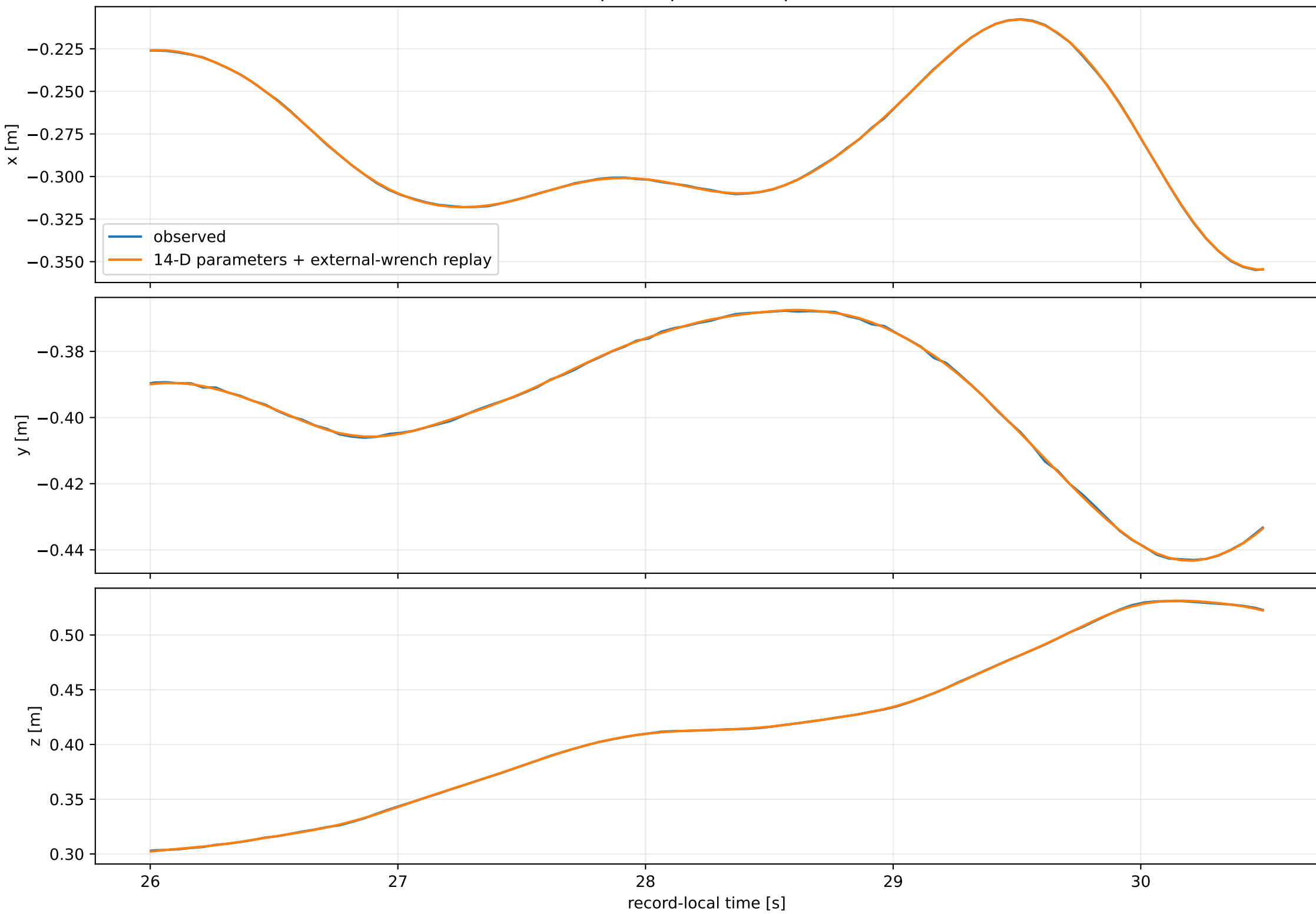
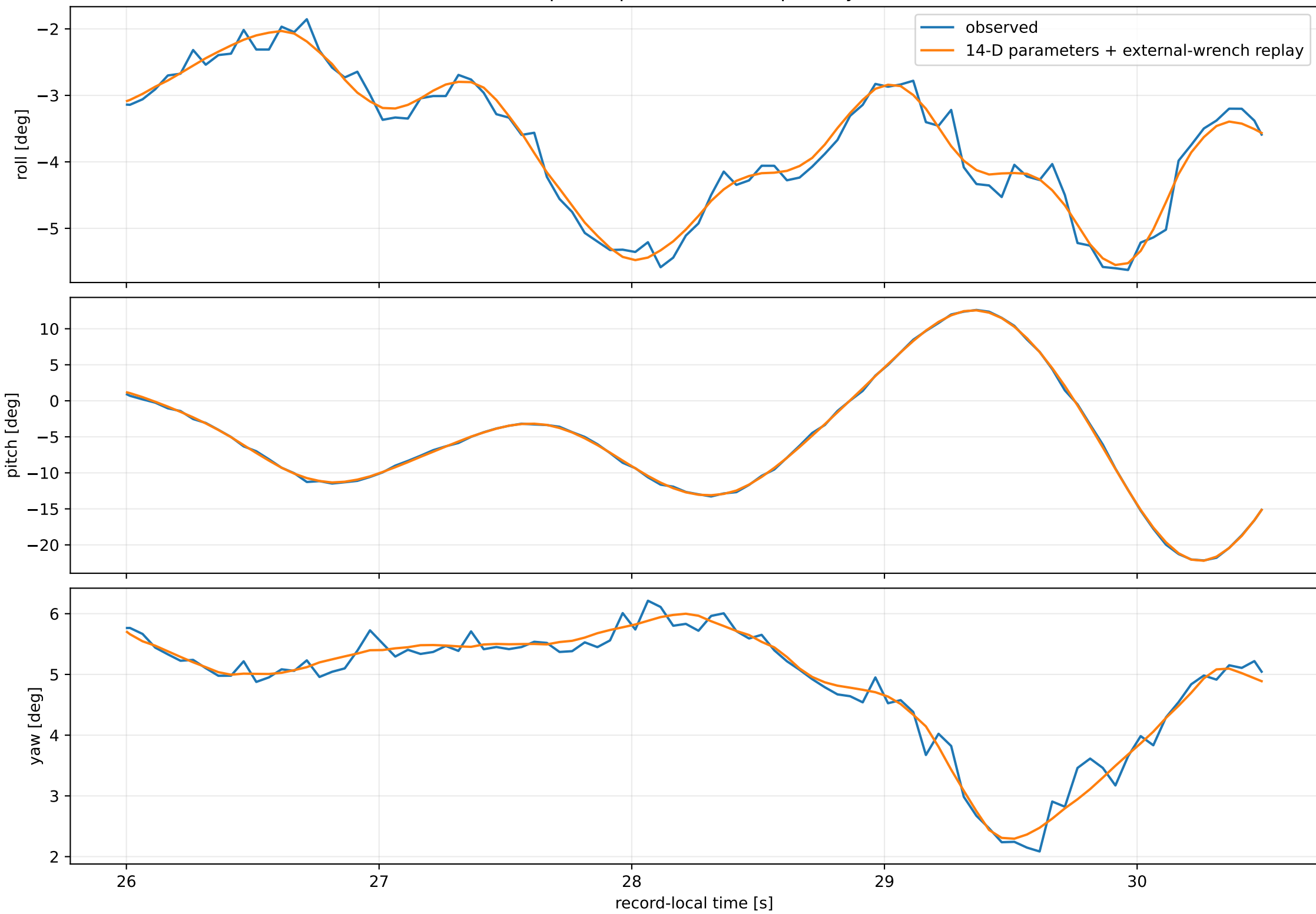


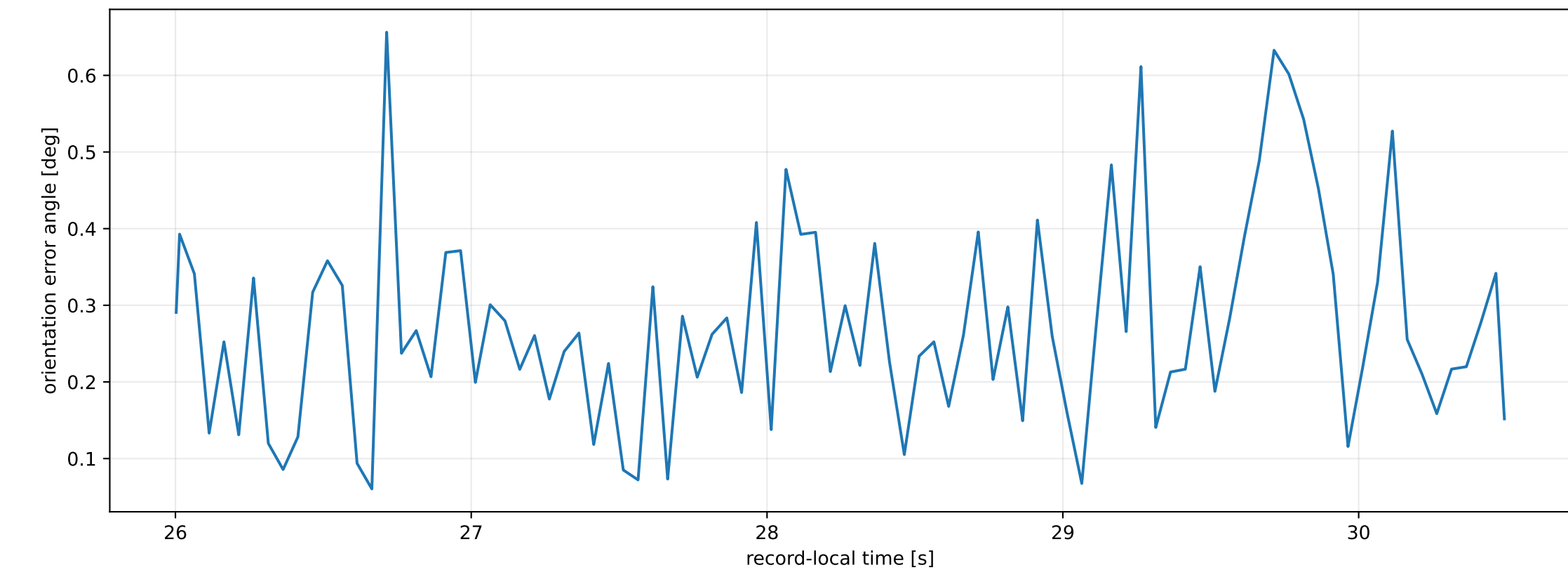
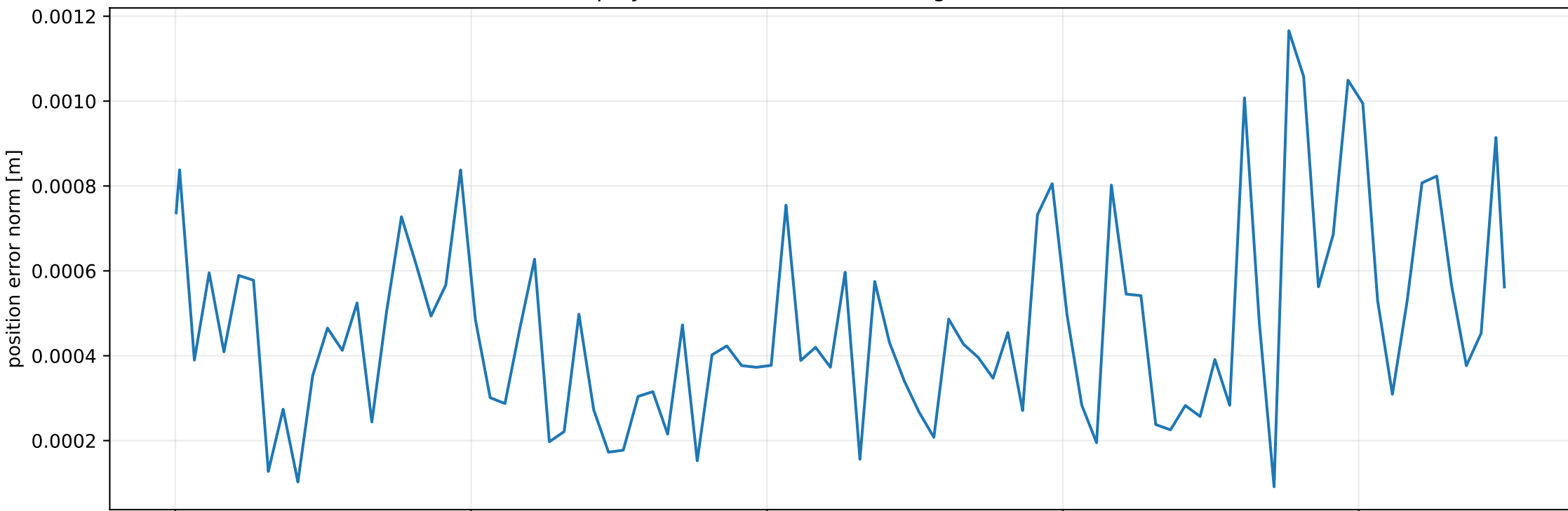
Fine pose reproduction: position



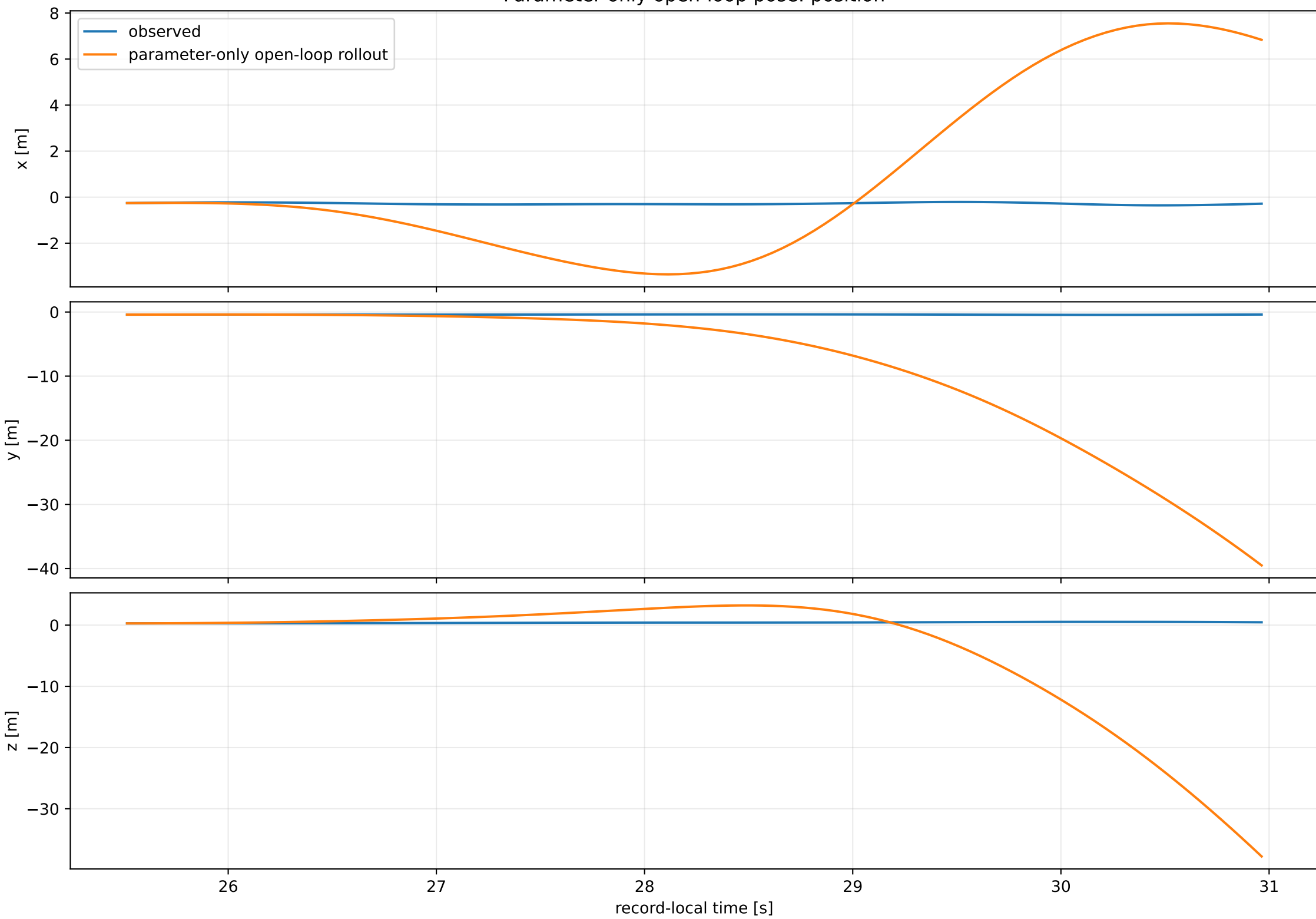
Fine pose reproduction: roll / pitch / yaw



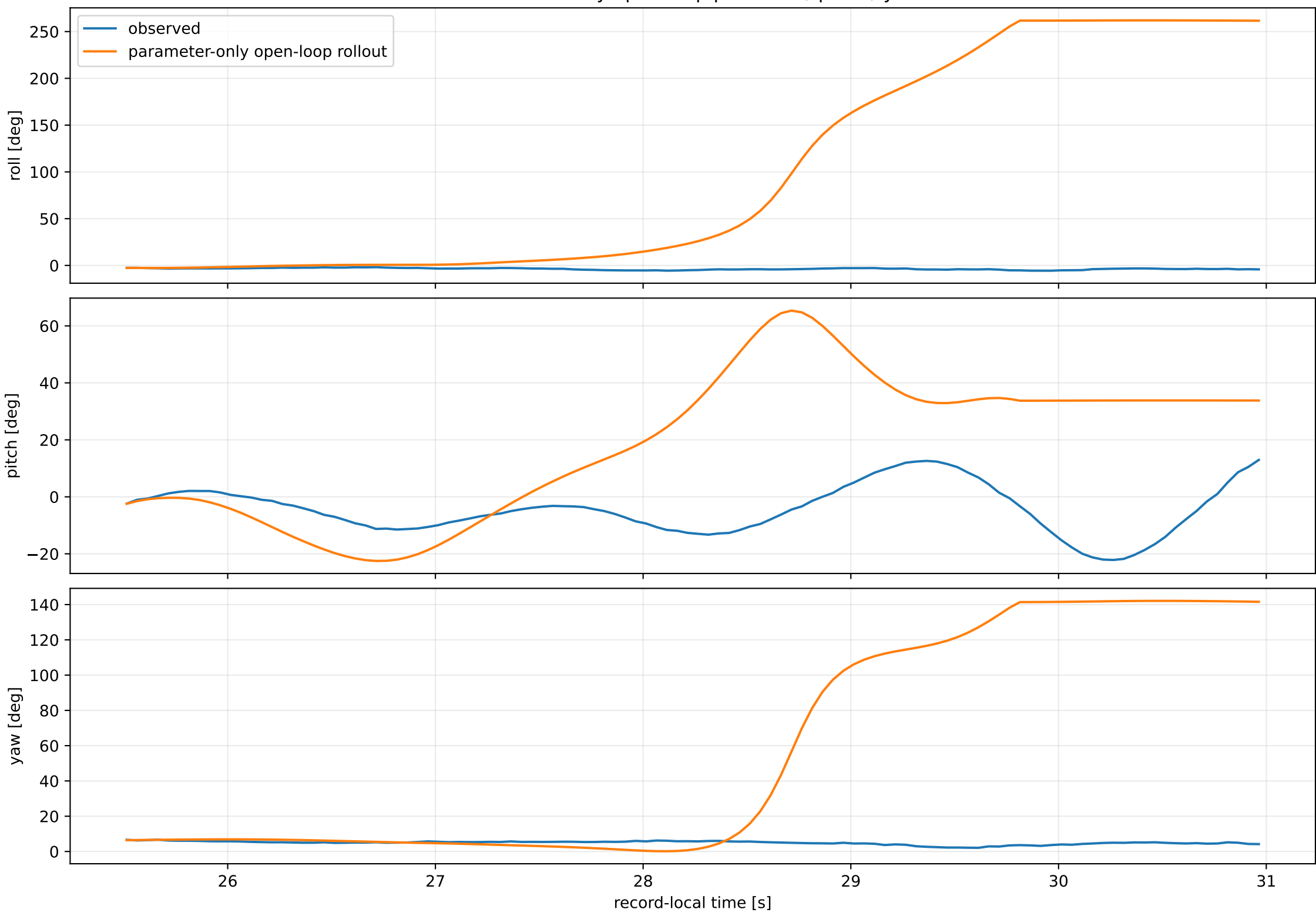
Fine replay error after the 14-D change — lower is better



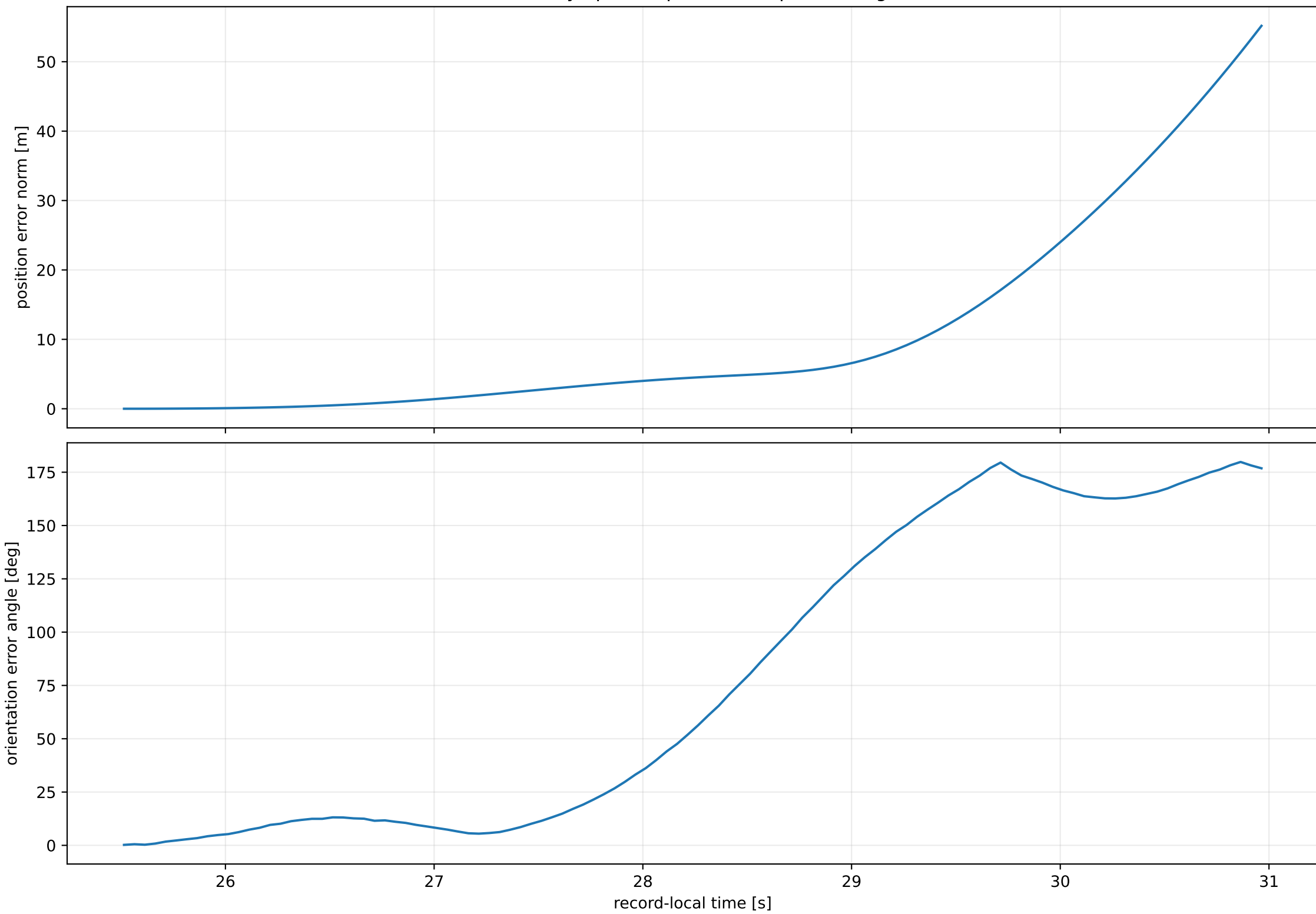
Parameter-only open-loop pose: position



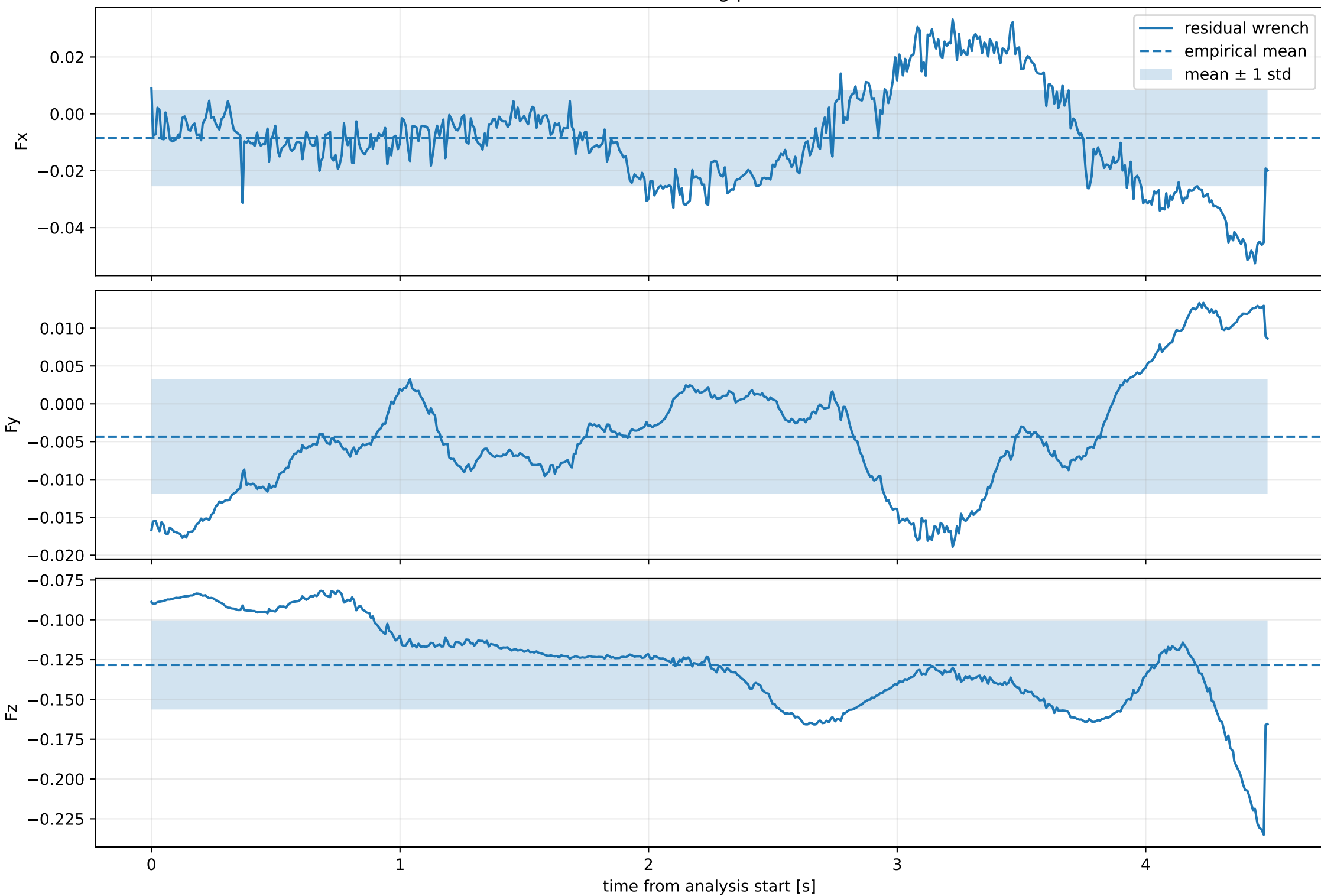
Parameter-only open-loop pose: roll / pitch / yaw



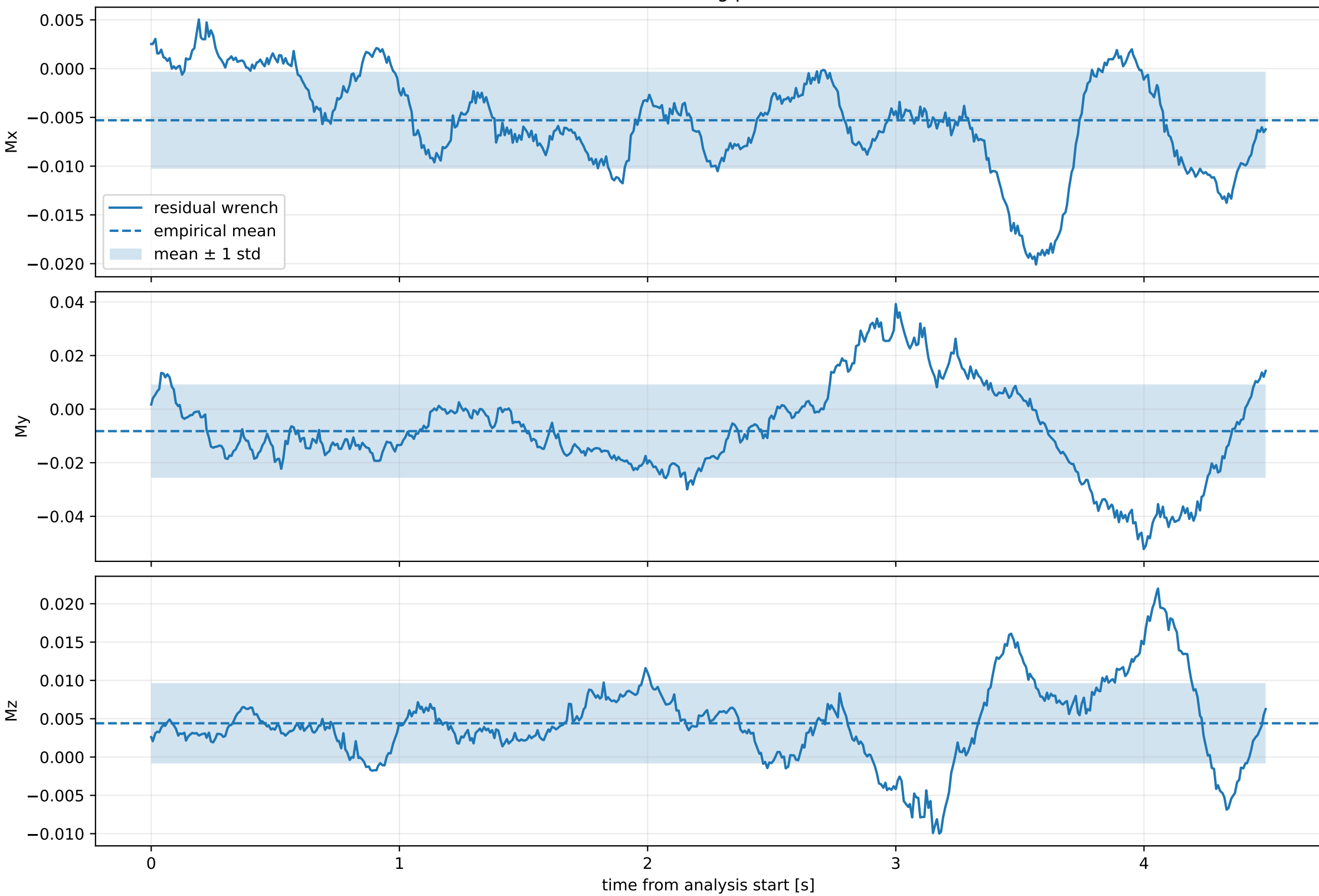
Parameter-only open-loop rollout (separate diagnostic)



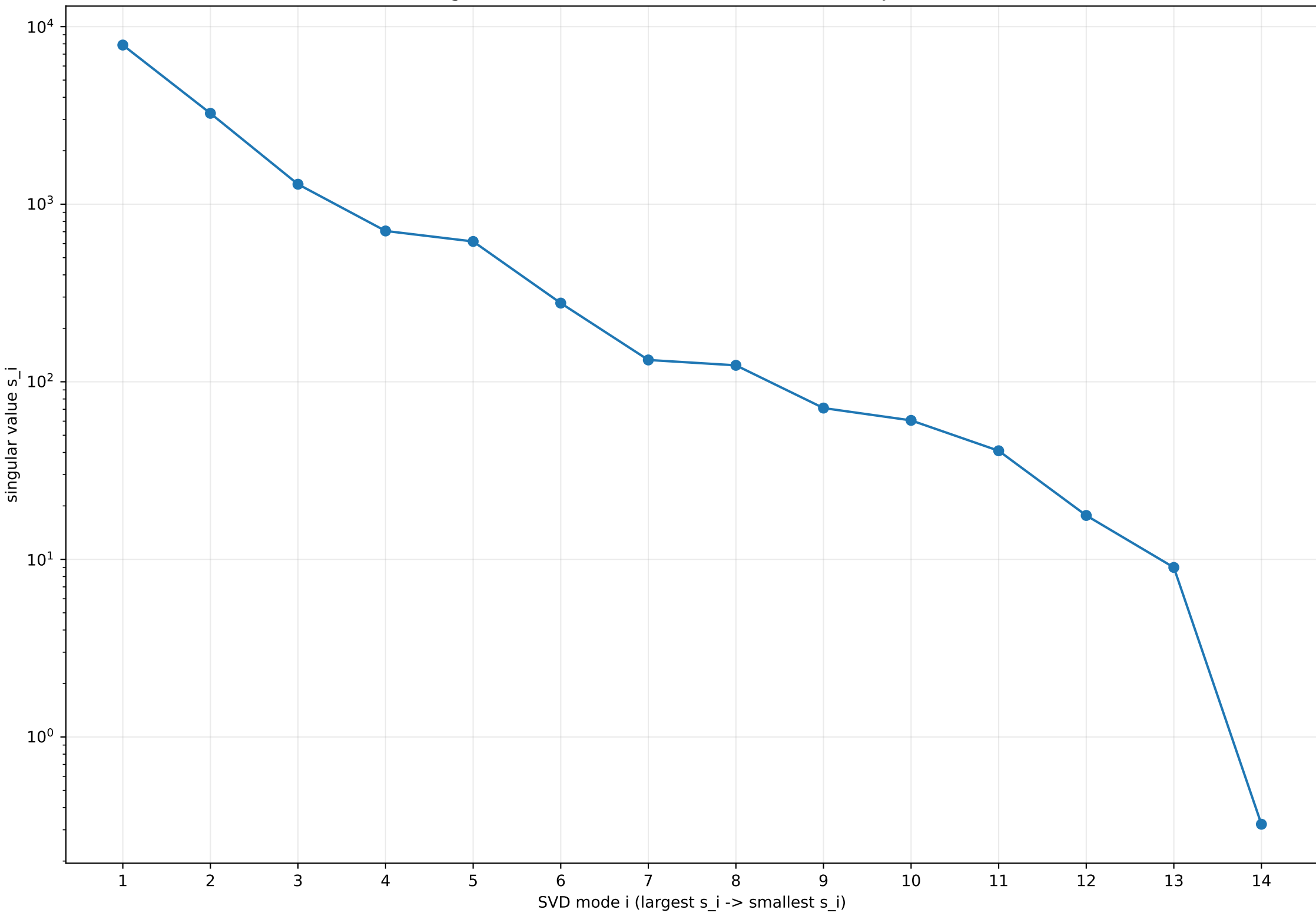
Dimensionless residual force: empirical mean ± 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



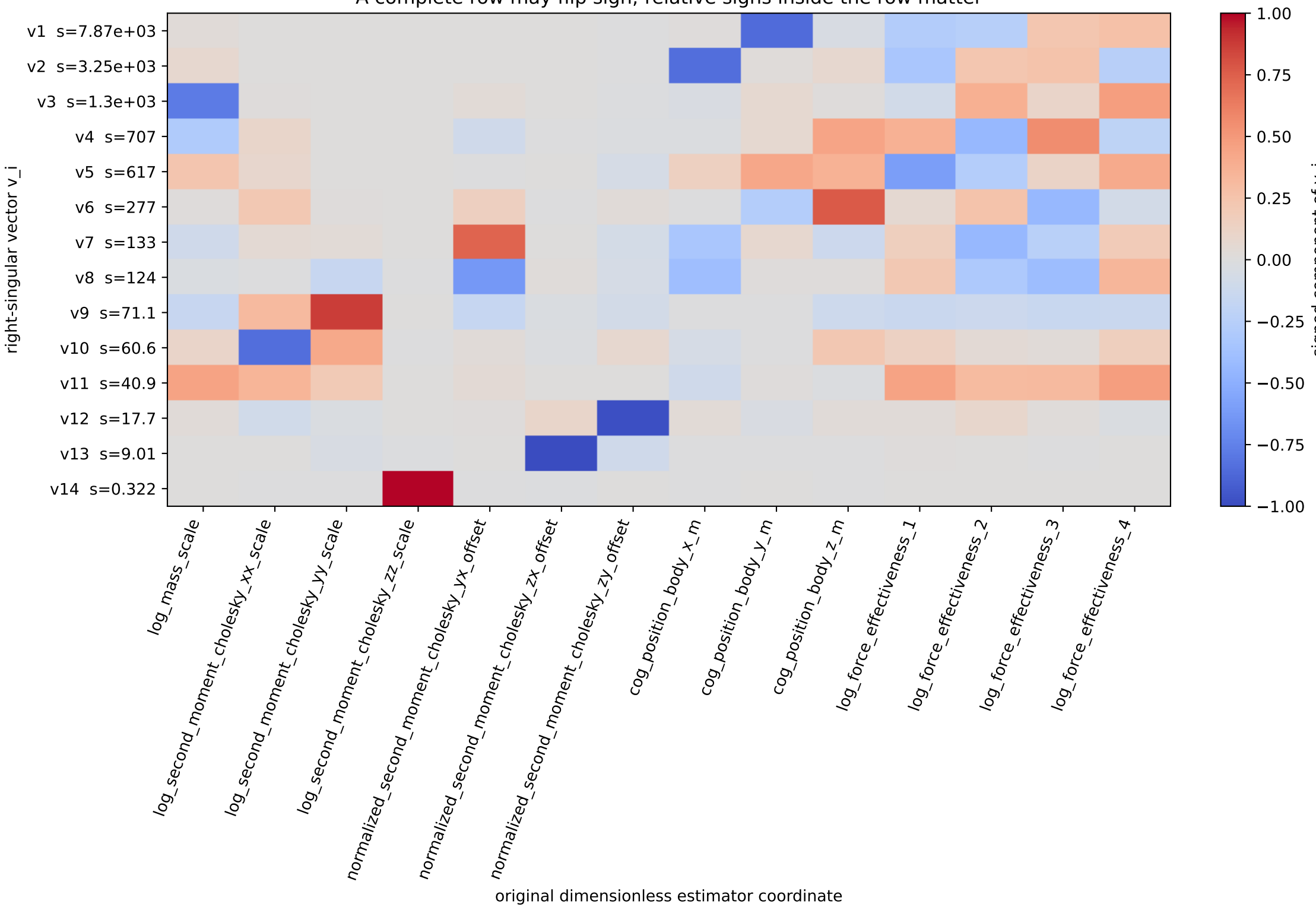
Dimensionless residual torque: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian



Signed right-singular vectors V^T
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.36630335395

inertia J [kg m²]

[0.065002075 5.1173861e-06 1.7407869e-05]
[5.1173861e-06 0.064981631 -2.9718326e-06]
[1.7407869e-05 -2.9718326e-06 0.12899244]

principal moments [kg m²]

[0.064980421, 0.065003281, 0.12899244]

CoG position in body frame [m]

[-0.0026394343, 0.00037730612, 0.0096938119]

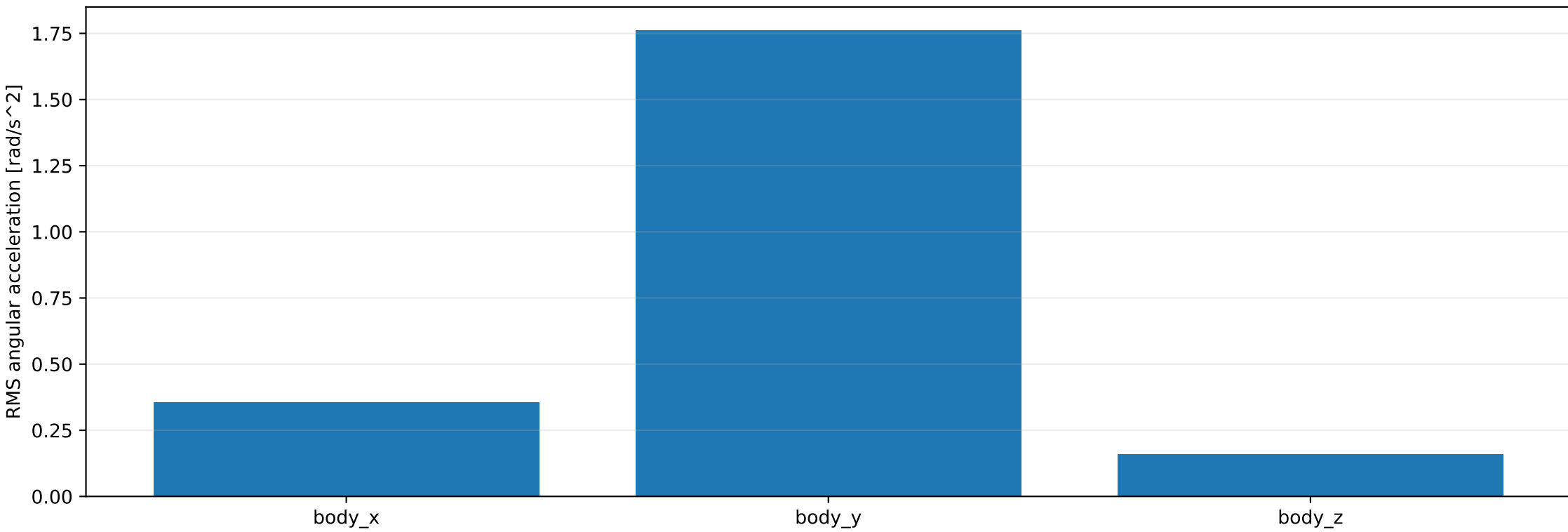
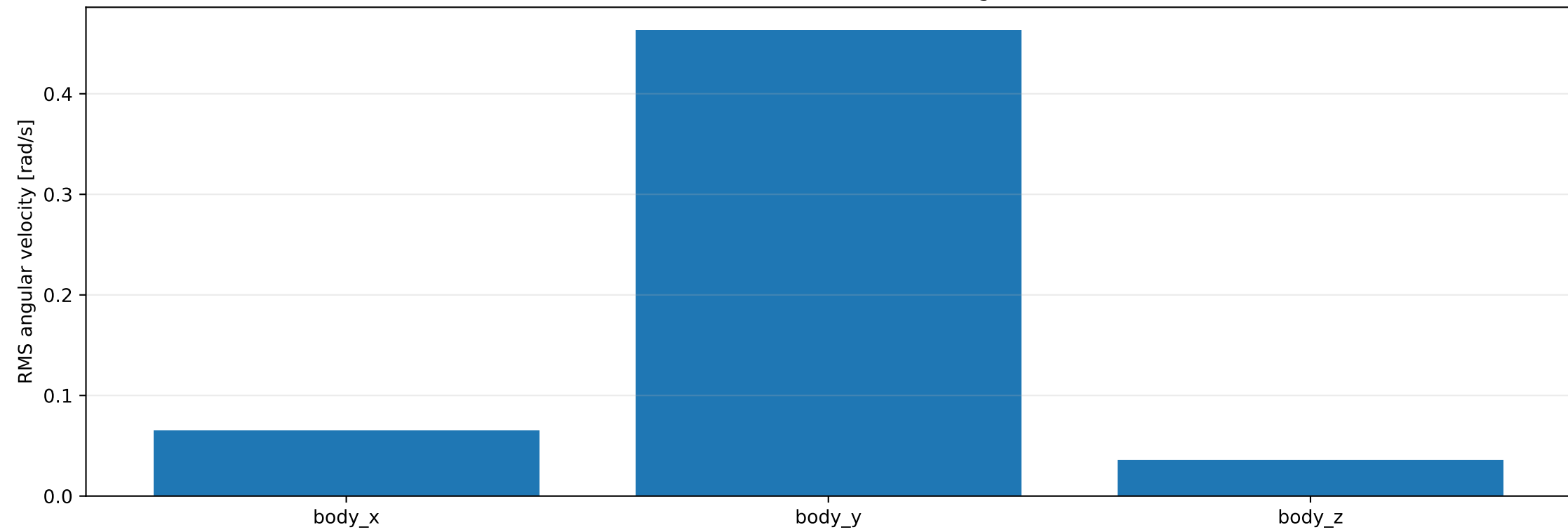
rotor force effectiveness

[0.96176936, 0.97108207, 0.96541865, 0.95631206]

selected delay [s]

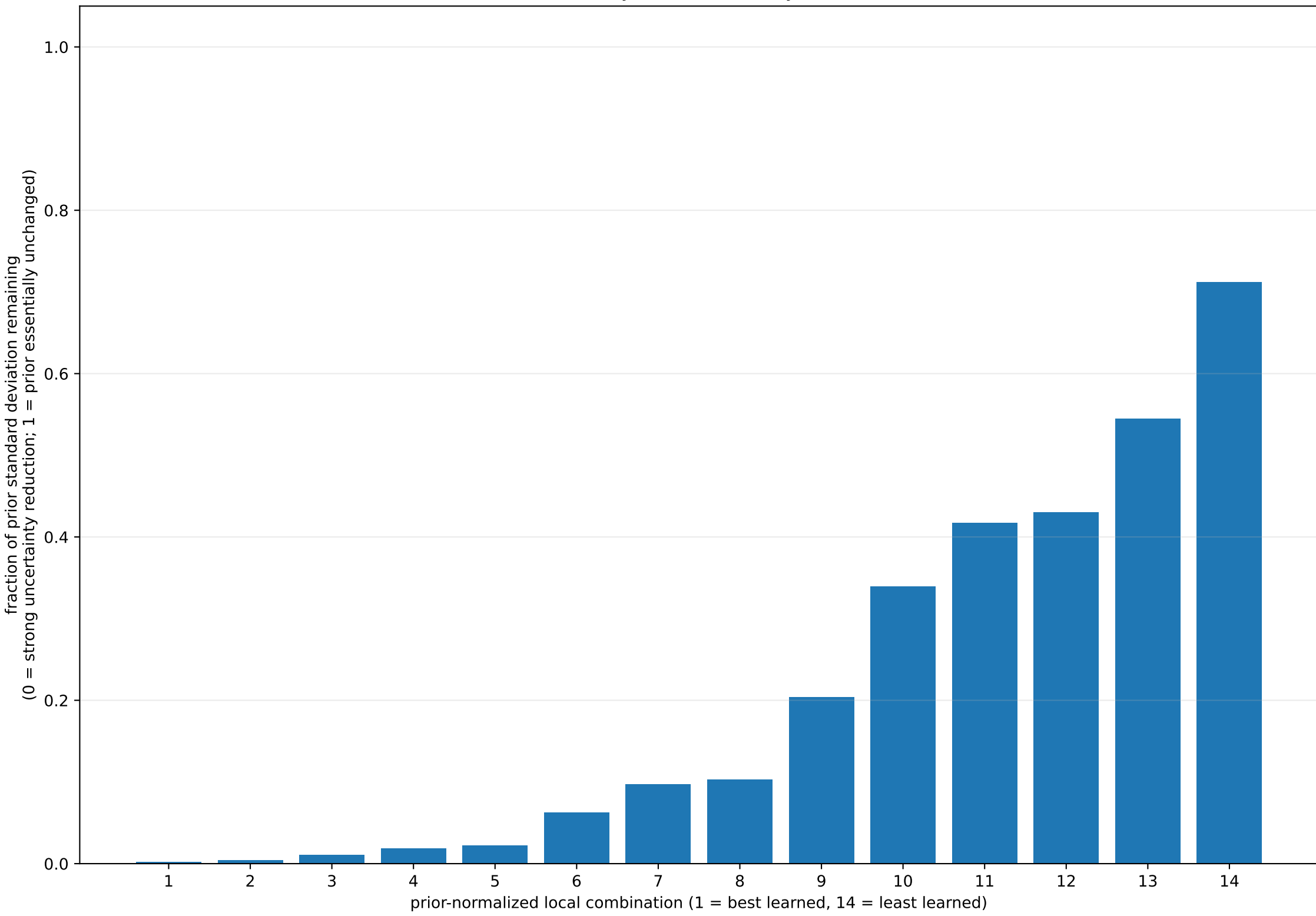
0.2

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

